

CURRICULUM VITAE

CONTACT

NAME **JUNGUK HUR**

INSTITUTE **School of Informatics, Indiana University, Bloomington**

COURSE **Master Degree in Bioinformatics**

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Homepage **<http://darwin.informatics.indiana.edu/juhur>**

EDUCATION

Aug. 2004 ~ Current **[INDIANA UNIVERSITY, School of Informatics](#)**
MS Degree in Bioinformatics

Aug. 2003 ~ Nov. 2003 **[S* Bioinformatics Online Course](#)**

Sep. 2001 ~ Dec. 2001 **[Strayer University, Maryland, USA](#)**
Computer Science Course – Introduction to Java Programming I.

June. 1, 2001 **[George Mason University, Prince William Campus, Virginia, USA](#)**
Computational Protein Structure Analysis and Structural Genomics

Jul. 1996 ~ Dec. 1996 **[University of New South Wales, Sydney, Australia](#)**
Exchange student at Department of Biochemistry and Molecular Biology

Jun. 1995 ~ Aug. 1995 **[English Language Program at UC Berkeley Extension, USA](#)**

Mar. 1994 ~ Feb. 2001 **[Pohang University of Science and Technology \(POSTECH\), KOREA](#)**
BS degree of Life Science. Graduation with honors (*cum laude*)
Thesis: ‘Impaired Olfactory Memory in Calcium Channel alpha 1B knockout mice’

WORKING EXPERIENCE

- Aug.2004 ~ Current Associate Instructor at Indiana University, School of Informatics
Fall 2005 (I519: Bioinformatics Theory and Application), **Spring 2005** (I692: Graduate Capstone Project, I590: Introduction to Molecular Biology and Genetics), **Fall 2004** (I692: Graduate Capstone Project)
- Jun. 2003 ~ Jun. 2004 *Part-time researcher at Bioinformatics team / [ISTECH Inc.](#)
Developed a Microarray analysis tool employing GOTM and participated in analysis processes of DNA Chip results via sequence annotation and biological analysis with literature information.
*Perl Programmer for mass analyses of nucleotide sequences with Ms. Yunkyung Lee at Yonsei University, Korea.
- Feb. 2001 ~ Nov. 2002 Researcher at Bioinformatics team / [ISTECH Inc.](#)
(USA Branch at Gaithersburg, MD and Head office in Korea)
Focused on functional analysis of genes from DNA chips and developed automatic functional annotation system using Perl scripts and sequence analysis tools. Categorized eukaryotic protein kinases and proteases and developed database ([IDPKP](#)). Managed development of '[GOODIES](#)', a biological interpretation program of Microarray clusters.
- Jul. 2000 ~ Dec. 2000 Lab participation / **National CRI Center for Calcium & Learning, Department of Life Science, POSTECH, KOREA**
Conducted researches on the possible roles of calcium channel alpha 1B by behavioral studies of knockout mice. Found that deficiency of normal CCα1b impairs the olfactory memory and the undergraduate thesis was prepared on this subject.
- Apr. 1998 ~ Jun. 2000 English linguist at **(MND) Military of National Defense Korea**
As an English interpreter for Korean and U.S. Army during compulsory military service
- Mar. 1997 ~ Aug. 1997 Lab participation / **Developmental Genetics Lab, Department of Life Science POSTECH, KOREA**
Assisted professor Hee-sup Shin in studying the roles of Phospholipase C isozymes that are selectively coupled to specific neurotransmitter receptors. Learned laboratory techniques such as DNA manipulation, PCR and etc.

PUBLICATION

- Sung geun Lee, **Jung Uk Hur**, and Yang suk Kim, (2004) A graph-theoretic modeling on GO space for biological interpretation of gene clusters. ***Bioinformatics.2004; 20: 381-388***

POSTER PRESENTATION

- James Costello, Mehmet Dalkilic, Rupali Patwardhan, **Junguk Hur**, Sumit Middha, Keval Mehta, Amit Saple, John Colbourne, Brian Eads, Justen Andrews. "InGeNE: Integrated Gene Network Explorer." i-Conference. Penn State University. **2005**
- Jungim Hur, Mona Damaj, Philip Beremand, Jenniffer Collins, **Junguk Hur**, Terry Thomas, (2004) The *Arabidopsis* ABI5 bZIP gene alters the global pattern of embryo-related and ABA-regulated gene expression. **American Society of Plant Biologists**, Lake Buena Vista, Florida USA
- **Jung Hur**, Sung Lee, Tae Chung, and Yang Kim, (2002) A Mathematical Modeling of GO Hierarchy for Gene Expression Profiling Analysis. **RECOMB 2002**. Washington D.C. USA.
- **Jung Hur**, Dong Park, and Yang Kim, (2002) Construction of Integrated Database for Protein Kinases and Proteases. **RECOMB 2002**. Washington D.C. USA.
- Sung Geun Lee, **Jung Uk Hur**, and Yang Suk Kim, (2002) GOODIES: A mathematical modeling of GO hierarchy for automated biological Validation and putative functional categorization of genes. **The 1st annual conference of the Korean society for bioinformatics**. Busan Korea.
- **Jung Uk Hur**, Wan Seon Lee, and Yang Suk Kim, (2002) Construction of Integrated Database for Protein Kinases and Proteases. **The 1st annual conference of the Korean society for bioinformatics**. Busan Korea.

PATENT

- A System for Analyzing DNA-Chips Using Gene Ontology, and a Method Thereof, 2002, Republic of Korea

SKILLS

- Fluent in Perl, Visual Basic .Net, HTML, CGI
- Moderate in MySQL, MATLAB, R Package
- Some in C++, JAVA

TOOLS

- PubChemSR(2005)http://darwin.informatics.indiana.edu/juhur/I571_ChemInformatics/index.html
- Sequence Alignment WebTools (2004) - <http://darwin.informatics.indiana.edu/juhur/tools.html>
- GOODIES (2001-3) - http://www.istech21.com/genom/genom_a03.html
- IDPKP (2001-3) - <http://www.istech.info/IDPKP/>

HONORS AND AWARDS CERTIFICATE

- Academic excellence awards 6 times at POSTECH, KOREA
- Graduation with honor (*cum laude*). BS in Life Science from POSTECH, KOREA
- Assistantship from School of Informatics, Indiana University, Bloomington.

RESEARCH INTERESTS

- Understanding quantitative transcriptional regulatory mechanism using ChIP-on-Chip data and related motif finding
- Functional genomics related sequence analyses
- Protein folding and structure prediction (Structural bioinformatics)